

Pakki Bhargava Shanmuka Sai

AI & Machine Learning Engineer | GitHub: <https://github.com/bhargav2313>

Mobile: +91 6302158512 Email: bhargavshanmukh2313@gmail.com LinkedIn: [linkedin.com/in/BhargavaShanmukaSai](https://www.linkedin.com/in/BhargavaShanmukaSai)

Summary

Seeking an entry-level position as a Machine Learning and AI Engineer to apply strong foundational knowledge in Python, data Science, and deep learning frameworks, contributing to the development of intelligent systems and innovative AI solutions that enhance business outcomes and operational efficiency.

Education

Bachelor of Technology in Computer Science and Engineering Lendi Institute of Engineering and Technology, Jonnada, Vizianagaram	August 2022 – May 2026 CGPA: 7.82
Board of Intermediate Education Sri Chaitanya Junior College, Vizianagaram	June 2020 – March 2022

Technical Skills

- **Programming Languages & GPU Computing:** Python, CUDA,C
- **Machine Learning & AI Frameworks:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain,OpenCv,YOLO,Pandas
- **API Development:** FastAPI, Streamlit
- **Computer Vision & NLP:** OpenCV, Natural Language Processing (NLP), Computer Vision, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Semantic Search
- **Robotics & Hardware:** Jetson Nano, Arduino, 3D Printing
- **Cloud Platforms & Services:** AWS, ECS, EKS, IAM, ECR
- **Version Control & CI/CD Tools:** Git, GitHub, GitHub Actions,Docker

Internships

Brainmage.ai AI/ML Intern	January 2025 – June 2025
------------------------------	--------------------------

- Engineered and optimized datasets by performing data preprocessing, cleansing, and EDA, ensuring high-quality inputs for training robust conversational AI models.
- Designed and implemented **(RAG)** pipelines for a Document Analysis Portal, integrating vector databases, embeddings,retrieval mechanisms, for improved contextual understanding in LLM-based applications.
- Collaborated with teams to implement scalable AI workflows, leveraging tools such as TensorFlow, PyTorch, and Hugging Face, ensuring reliable deployment of conversational AI prototypes.

SKOLAR Machine Learning with Data Science Intern	January 2024 – March 2024
---	---------------------------

- Completed intensive training in Machine Learning and Deep Learning, gaining hands-on experience with supervised/unsupervised learning, neural networks, CNNs, RNNs, NLP fundamentals, and model evaluation techniques.
- Built a minor project involving ETL pipelines for data extraction, transformation, and loading, applying data cleaning, feature engineering, and preprocessing techniques to prepare datasets for model training.
- Developed a major project focusing on Deep Learning and Natural Language Processing (NLP), implementing text preprocessing (tokenization, stemming, lemmatization), word embeddings (Word2Vec, GloVe), sentiment analysis, and sequence models (LSTM, GRU, Transformer-based architectures) for real-world text classification tasks.

Projects

AI-Document Chat &Comparison Portal	May 2025 – June 2025
-------------------------------------	----------------------

- Engineered an AI-powered document intelligence platform leveraging LangChain, RAG pipelines, vector databases (Chroma), semantic search, query rewriting (RAG Fusion) reranking (MMR), and multi-document Q&A capabilities.
- Implemented robust logging, exception handling, and integrated Groq inference along with Google API services for enhanced performance and external data processing
- Deployed full-stack application with HTML &CSS UI, FastAPI backend, Dockerized microservices, CI/CD pipelines (GitHub Actions), and AWS Fargate for scalable production-ready deployment..

Certifications & Trainings

- Certified in Fundamentals of Accelerated Computing with CUDA Python, C/C++ – NVIDIA.
- Achieved AWS Academy Graduate – Machine Learning Foundations.
- Completed AI Powered Data Analytics Mastery – Udemy.